

SaaS Tiffin Delivery Platform

Complete Project Flow Document

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1. What We Are Building

We are building a **SaaS (Software as a Service) platform** for the tiffin delivery industry. The idea is simple but powerful — any tiffin service business in India can sign up on our platform, get their own dashboard, their own WhatsApp bot, and their customers get a mobile app. Everything is managed from one central system that we control.

Think of it like Shopify, but for tiffin services. Shopify gives every shop their own online store. We give every tiffin service their own CRM, WhatsApp bot, delivery tracking, and customer app. The tiffin service owner does not need to worry about servers, apps, or technical setup — they just sign up, configure their menu, and start taking orders.

Why this matters

- Most tiffin services in India still run on phone calls, notebooks, and WhatsApp groups
- There is no affordable, ready-made CRM built specifically for this industry
- Our platform solves this by giving them a professional system at a low monthly cost
- We handle the tech, they focus on cooking and delivery

Business Model

We offer the platform on a subscription basis. Tiffin services (we call them **Admins**) pay a monthly or yearly fee to use the dashboard, WhatsApp bot, and app. We also charge a small markup on WhatsApp messages since we pay Meta for the API. Our target is to onboard the first 100 tiffin services.

2. The Four User Roles

The entire platform revolves around four types of users. Each one sees a different interface and has different permissions. Here is a quick overview before we go deep into each one.

Role	Interface	What They Do	Who They Are
Super Admin (Us)	Web Dashboard	Manages ALL tiffin providers on the platform. Handles billing, onboarding, support, analytics.	Platform owners (our team)
Admin (Tiffin Provider)	Web Dashboard	Manages THEIR customers, orders, menu, delivery partners, subscriptions, WhatsApp.	Each tiffin service owner
End User (Customer)	Mobile App + WhatsApp	Orders food, tracks delivery, manages subscriptions, makes payments.	People who order tiffins
Delivery Partner	Mobile App	Sees assigned deliveries, updates status, navigates to addresses.	Riders/drivers for each provider

Key Point: Every Admin (tiffin provider) is completely isolated from other Admins. Their customers, orders, delivery partners, and WhatsApp messages are all separate. One provider can never see another provider's data. This is called **multi-tenancy** — we will explain exactly how this works in Section 9.

3. Super Admin Panel (Web Dashboard)

The Super Admin panel is what **we** (the platform owners) use to manage the entire system. No tiffin service provider or customer ever sees this panel — it is exclusively for our internal team.

What the Super Admin Can Do

Provider Management

View all registered tiffin services, approve new signups, suspend accounts, manage trial periods, upgrade/downgrade plans

Billing & Payments

Track who has paid, send payment reminders, manage subscription plans (monthly/yearly), generate invoices

WhatsApp Management

Monitor WhatsApp API usage per provider, track message costs, manage phone number assignments, view delivery rates

Platform Analytics

Total providers, total end users, total orders across platform, revenue dashboard, growth metrics

Support & Tickets

Handle support requests from providers, escalate issues, manage FAQ content

Configuration

Set platform-wide settings, manage pricing plans, configure trial periods, set WhatsApp message markup rates

Provider Lifecycle (from Super Admin's View)

When a new tiffin service signs up, here is what happens on our end:

- 1. Sign Up:** Provider fills a registration form — business name, owner name, phone, city, area.
- 2. Trial Activation:** We activate a 14-day free trial. They get access to the dashboard, can add menu items, customers, etc.
- 3. WhatsApp Setup:** We register their WhatsApp Business number with Meta's API. This takes 1-2 days for verification.
- 4. Onboarding:** We help them upload their menu, import existing customers (if any), and configure delivery areas.
- 5. Go Live:** Their WhatsApp bot goes live. Their customers can start ordering via the app and WhatsApp.
- 6. Conversion:** Before the trial ends, we follow up to convert them to a paid plan.
- 7. Ongoing:** Monthly billing, usage monitoring, support as needed.

4. Admin Panel (Tiffin Service Provider Dashboard)

This is the main web dashboard that each tiffin service provider uses to run their business. Every provider gets their own isolated dashboard — they log in with their credentials and see only their own data.

Dashboard Modules

Home / Overview

- Today's orders count and status breakdown (pending, preparing, out for delivery, delivered)
- Revenue for today, this week, this month
- Active subscriptions count
- Low stock alerts
- Recent activity feed

Customers

- Full customer database with name, phone, email, area, address
- Customer tags for segmentation (e.g., 'Jain', 'No Onion', 'Bulk Order')
- Customer preferences and dietary notes
- Order history and lifetime value per customer
- One-click WhatsApp message to any customer

Orders

- Create new orders manually or receive them via WhatsApp/App
- Order status flow: Pending > Confirmed > Preparing > Out for Delivery > Delivered
- Assign delivery partner to each order
- Add customizations (extra roti, no spice, etc.)
- Delivery fee and total calculation
- Order event timeline for tracking

Menu / Products

- Add products with name, emoji, category, price, stock
- Stock tracking with low-stock alerts
- Stock movement history (who added/removed stock and when)
- Categories: Thali, Snacks, Beverages, Add-ons, etc.

Subscriptions

- Create recurring meal plans: Daily, Weekly, Bi-weekly, Monthly
- Pause and resume subscriptions
- Auto-generate orders based on subscription schedule

- Track next delivery date for each subscription

Delivery Partners

- Add riders with name, phone, area, vehicle type
- Mark partners as active/inactive
- Assign partners to orders
- Area-based auto-assignment (future feature)

WhatsApp Inbox

- View all WhatsApp conversations in one place
- Reply to customers directly from the dashboard
- See which messages are from existing customers vs. new leads
- Unread message count and notifications
- AI chatbot handles basic queries automatically

Leads Pipeline

- Track potential customers: New > Contacted > Interested > Converted / Lost
- Follow-up date reminders
- Convert a lead to customer with one click (auto-fills data)
- Source tracking: WhatsApp, Referral, Walk-in, Social Media

Reports & Analytics

- Revenue trends (daily, weekly, monthly charts)
- Top customers by order value
- Product performance (most ordered, revenue per product)
- Area intelligence (which areas bring most revenue)
- Cook plan (what to prepare today based on active orders and subscriptions)
- Customization analytics (most requested modifications)

Settings

- Business profile (name, address, logo)
- Backup and restore data
- WhatsApp bot configuration
- Notification preferences

5. End User Experience (Mobile App)

The end user is the person who orders tiffin — the customer of the tiffin service. They interact with two things: the **mobile app** and the **WhatsApp bot**. Let me explain both.

Mobile App (TiffinTrack)

The mobile app is built using Android (designed by Hussain using Google Stitch AI). The app is white-labeled — meaning each tiffin service's branding appears in the app, but the underlying technology is the same for all providers. When a customer opens the app, they see their specific tiffin service's menu, not anyone else's.

App Screens and Features

Home Dashboard

Shows today's meal status, delivery ETA, subscription summary, quick reorder button, and any announcements from the tiffin provider.

Meal Selection / Menu

Browse the full menu with categories (Thali, Snacks, Beverages). Each item shows name, photo, price, and an 'Add' button. Users can customize items (extra roti, no onion, etc.) before adding to cart.

Menu Item Details

Detailed view of a single item — description, nutritional info, allergen warnings, customer reviews, and customization options.

Cart & Checkout

Review selected items, apply promo codes, choose delivery address, select payment method, and place order.

Order Tracking

Real-time tracking of the current order — status updates (Confirmed, Preparing, Out for Delivery), delivery partner details, and estimated time.

Delivery History

Past orders with dates, items, amounts, and the option to reorder any previous meal with one tap.

Subscriptions

View active meal subscriptions, see upcoming deliveries on a calendar, pause/resume subscriptions, and change subscription plan.

Profile & Settings

Name, phone, email, saved addresses, payment methods, dietary preferences, notification settings.

Payments

Payment history, pending dues, wallet balance (if applicable), and payment method management. Supports UPI, cards, and cash on delivery.

WhatsApp Experience for End User

Not every customer will download the app. Many people, especially in smaller cities, prefer WhatsApp. So we have built a complete WhatsApp bot that can handle basic ordering and queries. Here is what a customer can do through WhatsApp:

- Say 'Hi' and get a welcome message with menu options
- Ask about the menu and see today's items with prices
- Place an order by typing item names and quantities
- Check their order status
- Ask about subscription plans
- Get a link to download the app for full features
- Report an issue or give feedback

6. Delivery Partner Flow

Each tiffin service has its own delivery partners — riders or drivers who deliver tiffins to customers. Here is how their flow works:

- 1. Registration:** The tiffin provider adds the delivery partner from the Admin dashboard — name, phone, area, vehicle type.
- 2. Assignment:** When an order is confirmed, the provider assigns a delivery partner from the dashboard. The partner gets a notification.
- 3. Pickup:** The delivery partner goes to the kitchen, picks up the orders assigned to them.
- 4. In Transit:** The partner marks orders as 'Out for Delivery' in their app. Customers can now see that their food is on the way.
- 5. Delivery:** Partner delivers to the customer's address. Marks order as 'Delivered'. Customer gets a notification.
- 6. Issues:** If there is a problem (wrong address, customer not available), the partner can mark it and the provider sees it in the dashboard.

Delivery Partner App Features

- See all assigned deliveries for today with customer name, address, and items
- Get optimized route for multiple deliveries
- Update delivery status in real-time
- View delivery history and earnings
- Call customer directly from the app (phone number masked for privacy)
- Area-based filtering — only see deliveries in their assigned area

7. WhatsApp Bot — How It Works Per Provider

This is one of the most important parts of our platform. Each tiffin service provider gets their own WhatsApp Business number with an AI-powered chatbot. Let me explain exactly how this works.

One Number Per Provider

Every tiffin provider who signs up on our platform gets their own WhatsApp Business number. This number is registered with Meta's Cloud API. When a customer messages this number, the bot responds on behalf of that specific tiffin service — with their menu, their prices, their delivery areas.

How the Bot Works (Step by Step)

Step 1 — Customer sends a message: For example: 'Hi, I want to order lunch'

Step 2 — Meta receives the message: Meta's WhatsApp Cloud API receives it and forwards it to our server via a webhook (a URL that Meta calls)

Step 3 — Our server identifies the provider: Based on the phone number the message was sent TO, our system knows which tiffin provider this belongs to. Every provider's phone number is mapped in our database.

Step 4 — Bot generates a reply: Our AI chatbot reads the message, checks the provider's menu, pricing, and delivery areas, and generates a helpful response.

Step 5 — Reply sent back: The reply is sent back through Twilio (our WhatsApp messaging partner) to the customer's WhatsApp.

Step 6 — Conversation is logged: The entire conversation is saved in the provider's WhatsApp inbox on their dashboard. The provider can see all chats and jump in manually if needed.

Bot Capabilities

Feature	How It Works	Example
Menu Inquiry	Bot pulls from provider's product list	"What's on the menu today?"
Place Order	Bot creates order in the system	"I want 2 thali and 1 juice"
Order Status	Bot checks order table	"Where is my order?"
Subscription	Bot shows plans and pricing	"Monthly plan details?"
Complaints	Bot logs issue, alerts provider	"My food was cold today"
App Download	Bot sends app link with provider code	"How do I download the app?"
Human Handoff	Bot flags chat for provider	"I want to talk to someone"

Cost of WhatsApp Messages

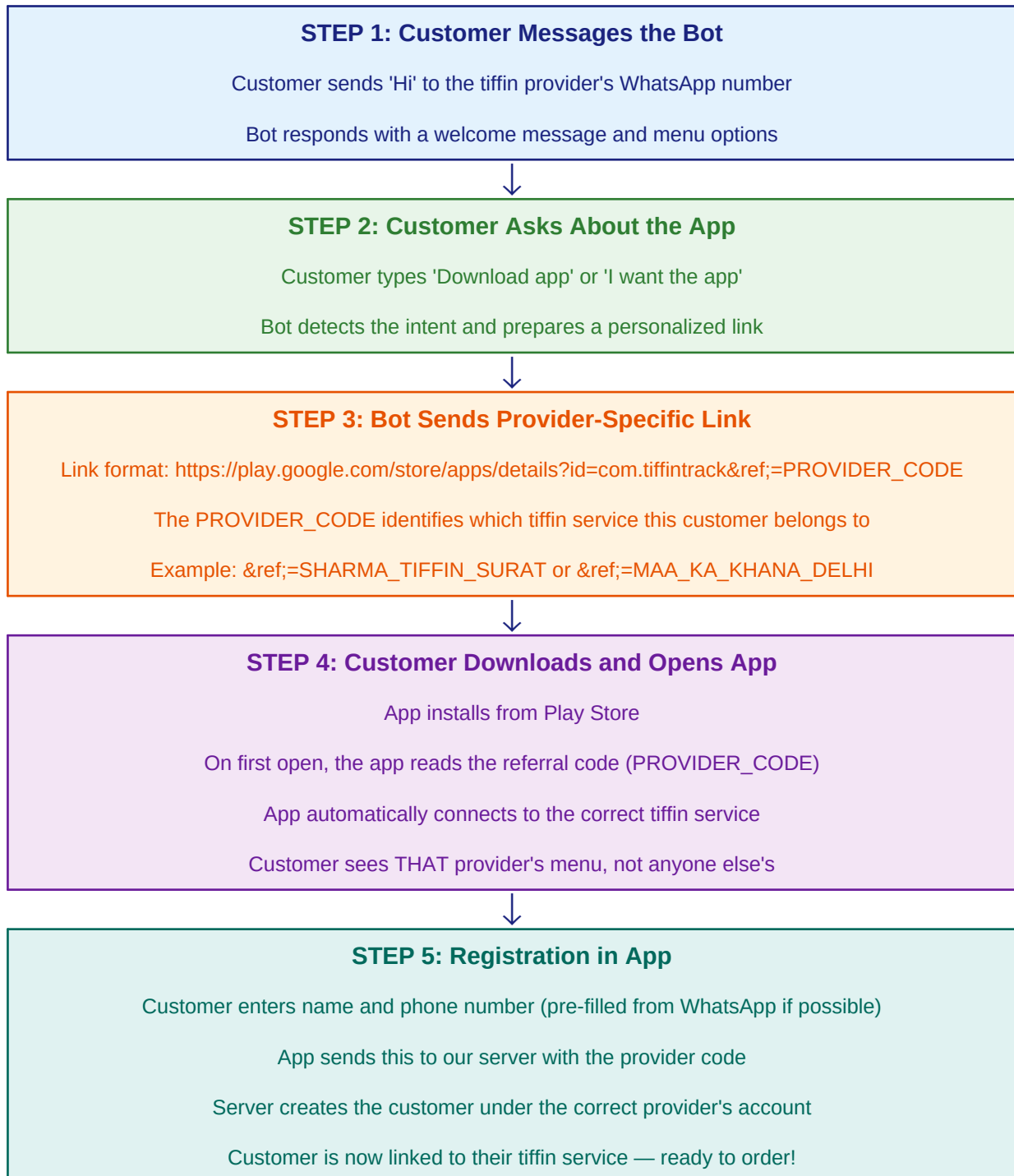
Meta charges per conversation (not per message). A conversation lasts 24 hours once initiated. Here are the current rates for India:

Category	Rate (INR)	When It's Used
Marketing	Rs 0.86	Promotions, offers, new menu announcements
Utility	Rs 0.12	Order confirmations, delivery updates
Authentication	Rs 0.12	OTP, login verification
Service	FREE	Customer-initiated conversations (first 1,000/month)

Important: Service conversations (customer messages first) are free for the first 1,000 per month. For a provider with 50-100 customers, most conversations will fall in this free tier. We add a small markup on top of Meta's pricing as our revenue.

8. How Users Download the App via WhatsApp

One of our smartest features is how we distribute the app. Instead of asking customers to search for our app on the Play Store (where they might get confused by similar apps), we use the WhatsApp bot to handle app distribution. Here is the complete flow:



Alternative Distribution Methods

- **QR Code:** Provider can print a QR code (generated from our dashboard) on their tiffin boxes, pamphlets, or shop counter. Scanning it opens the Play Store with their provider code embedded.

- **Direct Link:** Provider can share the download link on their social media, website, or in WhatsApp groups.
- **SMS:** Provider can send the app link via SMS to existing customers.

9. Multi-Tenant Architecture — Preventing Conflicts

This is the most critical technical section. Since multiple tiffin services use the same platform, we need to make absolutely sure that one provider's data never leaks to another. Here is how we achieve this.

What is Multi-Tenancy?

Multi-tenancy means one software system serves multiple 'tenants' (customers/businesses). In our case, each tiffin service provider is a tenant. They all use the same server, same app, same code — but their data is completely separated. Like an apartment building — everyone lives in the same building but has their own flat with their own lock.

How We Prevent Conflicts

Tenant ID on Every Record

Every single row in our database has a **tenant_id** field. This is the provider's unique ID. When a customer is created, their record includes `tenant_id`. When an order is placed, it includes `tenant_id`. When a WhatsApp message comes in, it is tagged with `tenant_id`. Every database query includes `WHERE tenant_id = ?` so a provider can only see their own data.

JWT Token Isolation

When a provider logs into the dashboard, they get a JWT (JSON Web Token) that contains their `tenant_id`. Every API call they make includes this token. The server extracts the `tenant_id` from the token and uses it to filter data. Even if someone tries to tamper with the request, the server validates the token and rejects unauthorized access.

WhatsApp Number Mapping

Each provider's WhatsApp Business number is stored in a mapping table: `phone_number > tenant_id`. When a message arrives on a specific number, the webhook handler looks up which provider it belongs to and routes the conversation accordingly. Two providers can never share the same WhatsApp number.

App Provider Code

When a customer downloads the app, the provider code (embedded in the download link) tells the app which provider to connect to. The app stores this code locally and sends it with every API call. The server validates that the customer belongs to the provider specified by the code.

Database-Level Separation

For the initial version (SQLite), each provider gets their own database file. Provider A's data is in `provider_a.db`, Provider B's is in `provider_b.db`. They literally cannot access each other's files. When we scale to Supabase/PostgreSQL in the future, we will use Row-Level Security (RLS) policies that enforce tenant isolation at the database engine level — even a buggy query cannot leak data.

API Middleware Enforcement

Every API endpoint passes through a middleware function that: (1) validates the JWT token, (2) extracts the `tenant_id`, (3) injects it into the database query. No endpoint can bypass this. Even if a developer forgets to add a tenant filter in a new endpoint, the middleware adds it automatically.

Conflict Scenarios and How We Handle Them

Scenario	Risk	Our Solution
Same customer orders from two different providers	Customer data overlap	Each provider has their own customer record. Same phone can exist under different tenants.
Provider tries to access another provider's data	Data breach	JWT token restricts access. API middleware enforces tenant_id.
WhatsApp message arrives for wrong provider	Wrong bot responds	Phone number mapping table ensures correct routing.
Delivery partner works for two providers	Order confusion	Partner records are per-tenant. Same person, two accounts.
Subscriber migration between providers	Data loss	Export/import tools available via Super Admin panel.

10. Technical Architecture Overview

Here is a simplified view of our technology stack and how all the pieces connect.

Technology Stack

Component	Technology	Purpose
Backend API	Node.js + Express.js	REST API server, handles all business logic
Database (Current)	SQLite (better-sqlite3)	Lightweight, file-based, zero cost, per-tenant DB files
Database (Future)	Supabase (PostgreSQL)	Managed cloud DB with Row-Level Security for multi-tenancy
Frontend (Admin)	Vanilla JavaScript	Web dashboard for providers and Super Admin
Mobile App	Android (Kotlin/Java)	Customer-facing app (TiffinTrack)
WhatsApp API	Meta Cloud API	Receives and sends WhatsApp messages
WhatsApp Delivery	Twilio	Sends outgoing WhatsApp messages reliably
AI Chatbot	Custom NLP Module	Understands customer messages and generates responses
Hosting	VPS / Cloud Server	Runs the Node.js backend and serves the frontend
Authentication	JWT (JSON Web Tokens)	Secure login for admin dashboard and app

API Structure

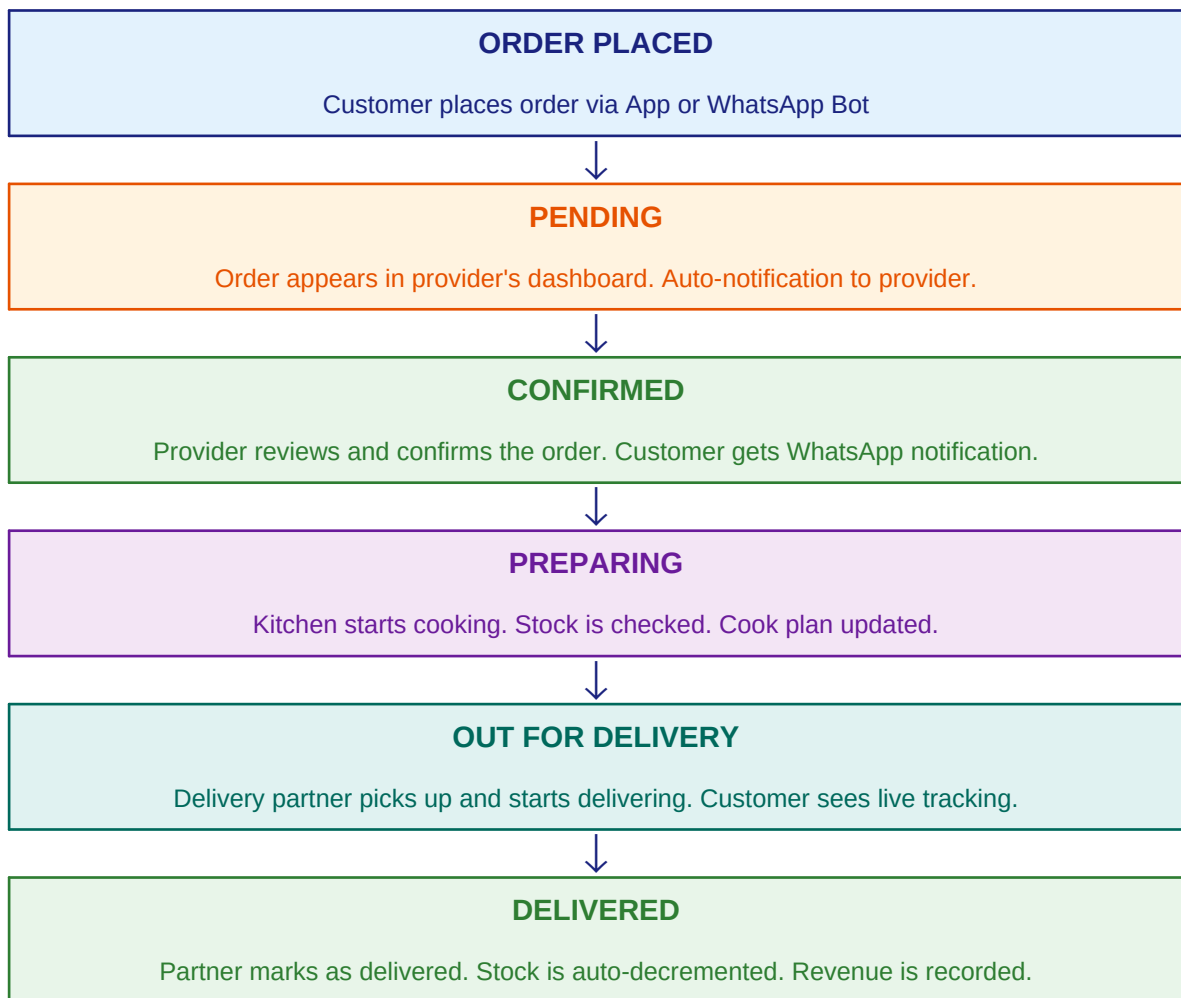
All API endpoints follow the pattern `/api/{resource}` and require JWT authentication (except webhooks):

Endpoint Group	Purpose	Key Endpoints
<code>/api/auth</code>	Login, user info, password	POST <code>/login</code> , GET <code>/me</code>
<code>/api/customers</code>	Customer CRUD	GET, POST, PUT, DELETE <code>/:id</code>
<code>/api/orders</code>	Order management	CRUD + PATCH <code>/status</code> , <code>/partner</code>
<code>/api/products</code>	Menu items + stock	CRUD + stock adjustments
<code>/api/subscriptions</code>	Meal plans	CRUD + PATCH <code>/toggle</code>
<code>/api/leads</code>	Sales pipeline	CRUD + POST <code>/convert</code>
<code>/api/partners</code>	Delivery partners	CRUD + toggle active
<code>/api/whatsapp</code>	WhatsApp integration	Webhook, threads, reply, mark-read

Endpoint Group	Purpose	Key Endpoints
/api/reports	Analytics	Summary, dashboard, cook-plan, area
/api/settings	Config + backup	GET/PUT settings, backup/restore

11. Complete Order Lifecycle

An order goes through several stages from the moment a customer places it to when it is delivered. Here is the complete lifecycle with who does what at each stage.



What Happens at Each Stage (Technical)

Pending: Order record created in database with status='pending'. Activity log entry added. If from WhatsApp, bot confirms receipt.

Confirmed: Provider clicks 'Confirm' on dashboard. Status updated to 'confirmed'. Event added to order timeline. WhatsApp utility message sent to customer (cost: Rs 0.12).

Preparing: Provider moves to 'preparing'. Cook plan on dashboard updates automatically to show what needs to be cooked.

Out for Delivery: Delivery partner assigned. Partner's app shows the delivery details. Customer's app shows tracking. Status = 'out_for_delivery'.

Delivered: Partner marks delivered. Stock for each item auto-decremented. Order total added to revenue. Activity log updated. Customer can rate the meal in the app.

Cancelled: If cancelled at any point, stock is not decremented. Reason is logged. Customer notified via WhatsApp.

12. Mobile App Screens (TiffinTrack Design)

The mobile app was designed by Hussain using Google Stitch AI. Below is a description of each screen and how it connects to the backend system. The design follows Material Design principles and is optimized for easy one-handed use.

Home Dashboard

- Greeting banner with user's name and today's date
- Today's meal card — shows what's being delivered today (for subscription users)
- Quick action buttons: Order Now, View Menu, Track Delivery, My Subscription
- Recent order status widget (shows last order with live status)
- Announcements/offers from the tiffin provider (holiday closures, new items, discounts)
- Bottom navigation: Home, Menu, Orders, Subscription, Profile

Meal Selection / Menu

- Category tabs: All, Thali, Snacks, Beverages, Add-ons
- Grid/list view toggle
- Each item card: image, name, price, rating, 'Add to Cart' button
- Search and filter (vegetarian, non-veg, Jain, etc.)
- Cart preview floating button with item count

Menu Item Details

- Large item image
- Name, description, and price
- Nutritional information (if available)
- Customization options: checkboxes for 'Extra Roti', 'Less Spice', 'No Onion', etc.
- Quantity selector
- Customer reviews and ratings
- 'Add to Cart' button at the bottom

Delivery History

- List of past orders sorted by date (newest first)
- Each order shows: date, items, total amount, delivery status
- Tap to see full order details with timeline
- 'Reorder' button on each past order for quick repeat ordering
- Filter by date range, status

Profile & Settings

- Profile picture, name, phone number

- Saved addresses (home, office, other)
- Dietary preferences: Veg, Non-Veg, Jain, Vegan
- Notification preferences
- Payment methods management
- App language selection (Hindi, English, Gujarati)
- Help and Support link
- Logout

Payments

- Current balance / pending dues
- Payment history with dates and amounts
- Pay now button (UPI, card, wallet)
- Auto-pay setup for subscriptions
- Download payment receipts
- Cash on Delivery option toggle

13. Data Flow and Security

Security is not an afterthought — it is built into every layer of the system. Here is how we protect data and ensure privacy.

Authentication: Every user (admin, customer, delivery partner) logs in with credentials and receives a JWT token. The token expires after 24 hours. All API calls require a valid token.

Password Security: Passwords are hashed using bcrypt before storing in the database. We never store plain text passwords. Even we cannot see a user's password.

HTTPS Everywhere: All communication between the app/browser and our server is encrypted using HTTPS (SSL/TLS). No data travels in plain text.

Tenant Isolation: As explained in Section 9, every data operation is filtered by `tenant_id`. The middleware enforces this at the API layer, and the database enforces it at the storage layer.

WhatsApp Encryption: WhatsApp messages are end-to-end encrypted by default. We only store the message content that Meta forwards to us via webhook — we never see the encryption keys.

Backups: Each provider can download a complete backup of their data as JSON. The Super Admin can also take platform-wide backups. Backups are encrypted at rest.

Access Control: Super Admin, Admin, End User, and Delivery Partner each have different permission levels. A delivery partner cannot access customer payment data. A customer cannot access another customer's data.

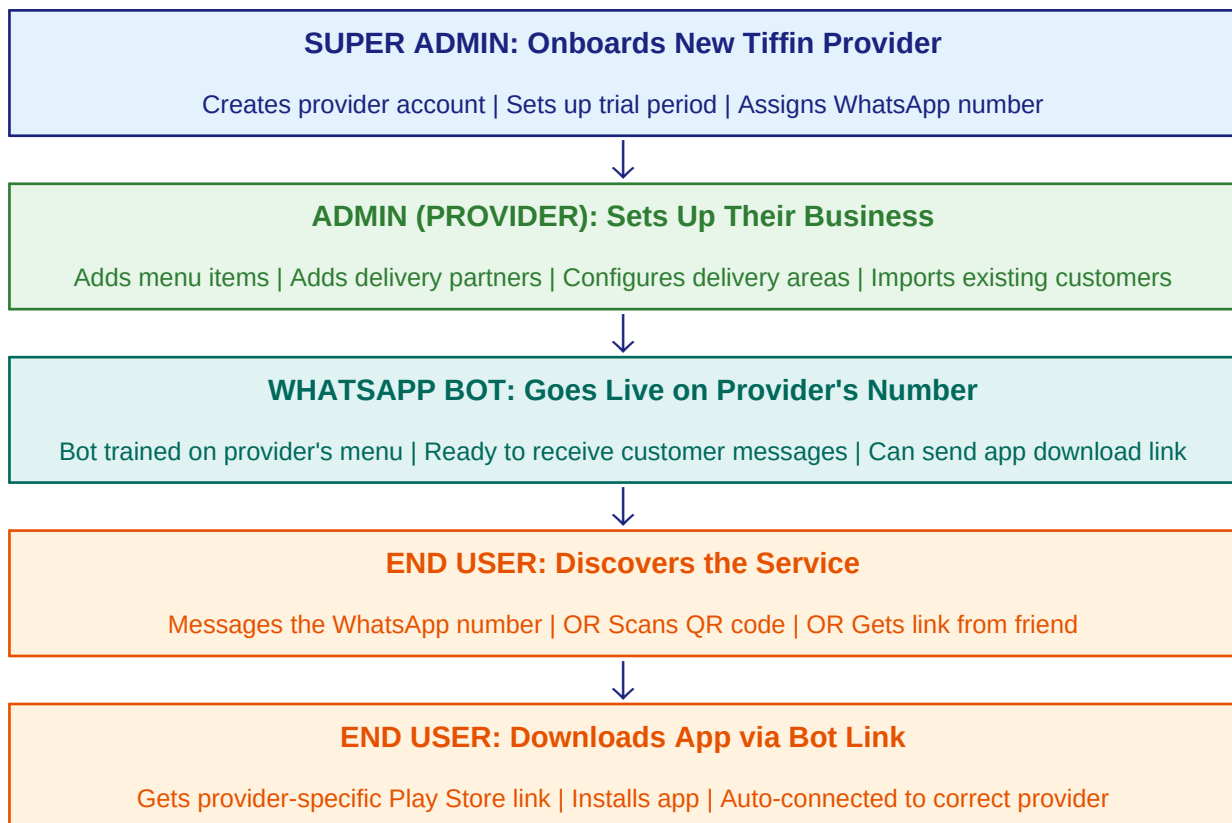
API Rate Limiting: To prevent abuse, we limit the number of API calls per minute. This protects against bots, scrapers, and denial-of-service attacks.

14. Master Flowchart — Everything Together

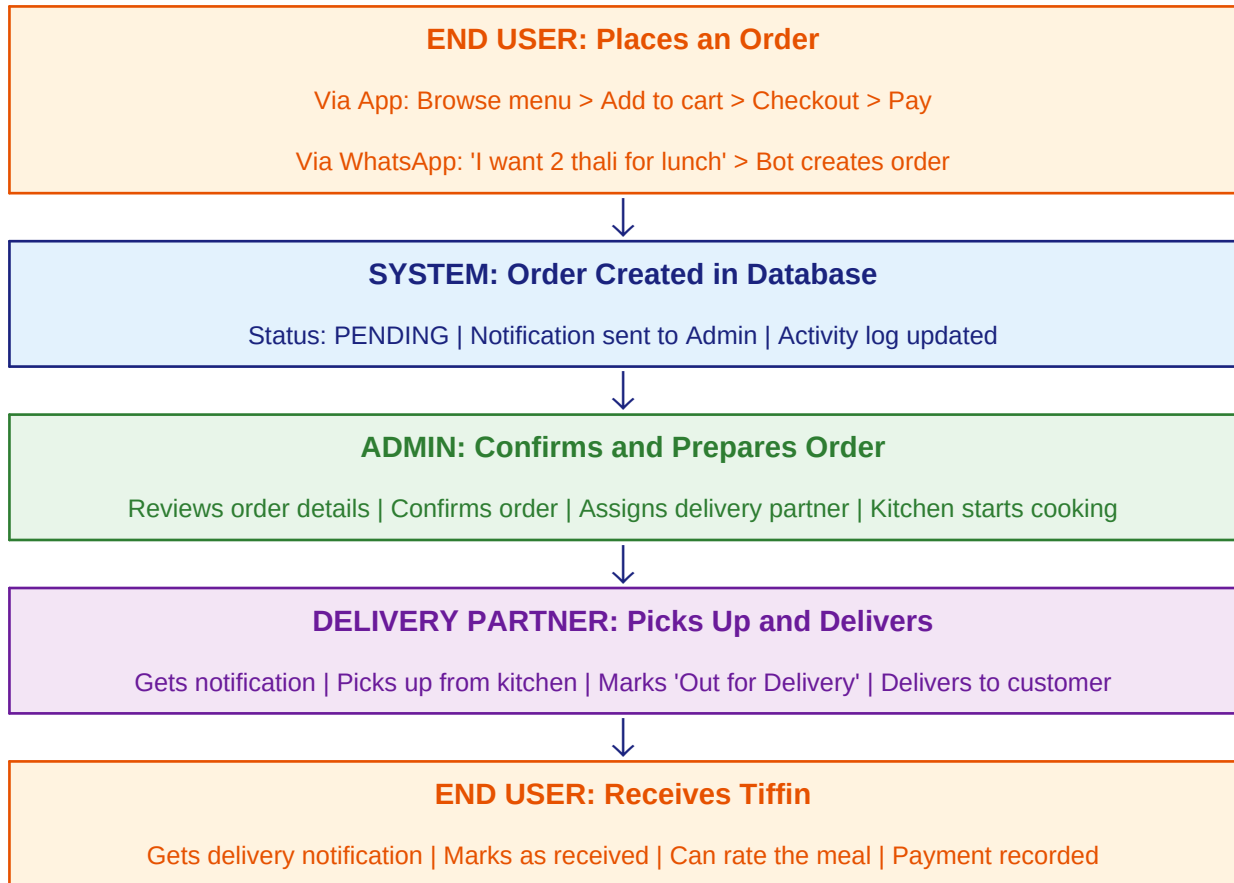
This section ties everything together into one visual flow. Each color represents a different user role, and the arrows show how data and actions flow between them.

Color	Role	Description
Blue	Super Admin	Platform management, billing, provider onboarding
Green	Admin (Provider)	Daily operations, orders, menu, WhatsApp inbox
Orange	End User	Ordering food, tracking delivery, managing subscription
Purple	Delivery Partner	Picking up and delivering orders
Teal	WhatsApp Bot	Automated customer interaction per provider

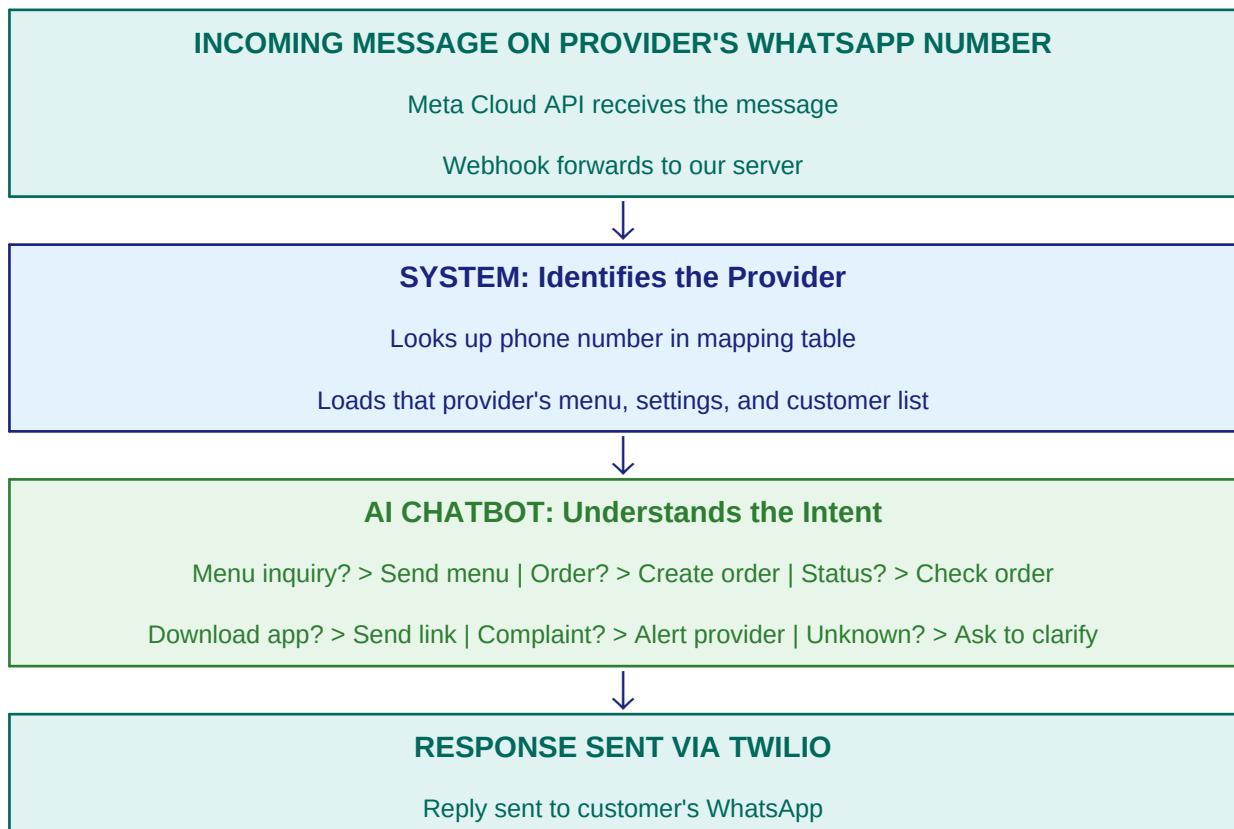
Platform Setup Flow



Daily Operations Flow



WhatsApp Bot Decision Flow



Conversation logged in provider's inbox | Provider can take over anytime

Complete End-to-End Flowchart

Below is the full system flow showing how all four roles interact from start to finish:

Stage	Super Admin	Admin (Provider)	End User	Delivery Partner
1. Onboarding	Creates provider account, assigns plan	Receives credentials, logs into dashboard	-	-
2. Setup	Registers WhatsApp number with Meta	Adds menu, partners, delivery areas	-	Added by provider, gets app access
3. Discovery	Monitors platform metrics	Shares WhatsApp number, QR codes	Messages WhatsApp or scans QR	-
4. App Download	-	Bot sends provider-specific app link	Downloads app, auto-linked to provider	-
5. Ordering	-	Receives order on dashboard	Places order via app or WhatsApp	-
6. Preparation	-	Confirms order, starts cooking	Gets confirmation notification	-
7. Dispatch	-	Assigns delivery partner	Sees 'Out for Delivery' status	Gets assignment, picks up order
8. Delivery	-	Sees delivered in dashboard	Receives tiffin, rates meal	Delivers, marks complete
9. Billing	Tracks provider payments	Pays platform subscription	Pays for meals (UPI/Cash/Card)	Earnings tracked by provider
10. Growth	Converts trial to paid plans	Uses reports to optimize business	Subscribes for daily meals	Covers more areas

Summary

This document covers the entire flow of our SaaS tiffin delivery platform — from how a provider signs up, to how their customers order food, to how delivery happens, to how we prevent data conflicts between providers. The key takeaways are:

- **Four distinct user roles** — each with their own interface and permissions
- **WhatsApp bot per provider** — handles ordering, queries, and app distribution
- **Smart app distribution** — provider-specific links ensure customers connect to the right service
- **Multi-tenant isolation** — `tenant_id` on every record, JWT enforcement, separate databases
- **Complete order lifecycle** — from placement to delivery with real-time tracking
- **Scalable architecture** — starts with SQLite, ready to migrate to Supabase when we grow

